

TEEM-CMB 2.0

A Tension-Field Interpretation of CMB Anisotropies Beyond the First Acoustic Peak

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Abstract

We present TEEM-CMB 2.0, an extension of cosmic microwave background (CMB) physics based on the Tension–Energy Exchange Model (TEEM). In this framework, spacetime consists of three interacting energetic modes—vibrational energy v , condensed mass-energy m , and spatial tension R . We show that CMB anisotropies arise not only from photon temperature fluctuations but also from fluctuations in the tension field, leading to the combined observable

$$\delta T_{\text{obs}} = \delta T_{\gamma} + f_R \delta R.$$

A key ingredient is the **partial reset**, an early-Universe energetic transition that suppresses high-frequency vibrational modes and imprints a primordial spectrum of tension fluctuations $P_R(k)$. These fluctuations contribute directly to the gravitational potential and modify the effective flow of time through the ratio v/R , altering both the amplitude and phase of acoustic oscillations. As a result, TEEM predicts distinctive signatures in the CMB power spectrum, including suppressed odd peaks, enhanced even peaks, a smoother damping tail, reduced early-ISW power, and large-scale anisotropies such as hemispherical asymmetry and parity violation. Because the tension field modifies both the sound horizon and the angular-diameter distance, the position of the first acoustic peak ($\ell_1 \approx 220$) is **not** preserved and cannot serve as a consistency test for TEEM.

The primordial tension spectrum generated during the partial reset maps directly onto the galactic tension-coherence scale r_b in TEEM-DM 2.0, establishing a unified energetic origin for both CMB anisotropies and galactic rotation-curve diversity. TEEM-CMB 2.0 remains fully consistent with existing observations while providing natural explanations for several Λ CDM anomalies. Upcoming experiments—including the Simons Observatory, CMB-S4, and LiteBIRD—will be capable of testing the key predictions of the tension-field contribution to the CMB.

1. Introduction

The cosmic microwave background (CMB) provides the earliest directly observable imprint of the Universe. In the standard Λ CDM framework, its anisotropies are interpreted almost entirely as **photon temperature fluctuations** generated by acoustic oscillations in the primordial plasma. This interpretation has been remarkably successful in explaining the overall structure of the angular power spectrum, yet several persistent anomalies remain unresolved:

- the low- ℓ power deficit,
- the hemispherical asymmetry,

- the parity anomaly,
- the Cold Spot,
- and small but significant deviations in the damping tail.

These features challenge the assumption that **temperature fluctuations alone** encode the full physical content of the CMB.

The Tension–Energy Exchange Model (TEEM) offers a different perspective. In TEEM, spacetime is an energetic medium composed of three interacting modes—vibrational energy ν , condensed mass-energy m , and spatial tension R . While ν and m evolve rapidly in the early Universe, the tension field R evolves slowly and retains long-range coherence. TEEM-CMB proposes that **the CMB anisotropies arise not only from photon temperature fluctuations δT , but also from fluctuations in the tension field $\delta R(z)$** , inherited from the partial-reset epoch of the early Universe.

This leads to a fundamental reinterpretation of the observed CMB signal:

$$\delta T_{\text{obs}}(\hat{n}) = \delta T_{\gamma}(\hat{n}) + f_R \delta R(z, \hat{n}),$$

where f_R is a coupling determined by TEEM dynamics. In this view, the CMB encodes both thermal and tension-field information, and the anisotropies reflect the combined energetic structure of spacetime at recombination.

A key implication follows immediately: **the position of the first acoustic peak ($\ell_1 \approx 220$) is not a robust test of TEEM**. In Λ CDM, ℓ_1 is tightly linked to a fixed sound horizon and a fixed angular-diameter distance. In TEEM, however, the effective “ruler” used in this projection is modified by the redshift-dependent tension field $R(z)$. Because $R(z)$ alters the mapping between physical and angular scales, **the peak position is expected to shift**, and matching $\ell_1 = 220$ is neither required nor physically meaningful within TEEM.

Instead, TEEM-CMB predicts that **peak height ratios, damping-tail behavior, early-ISW signatures, and large-scale anomalies** provide the correct observational tests. These quantities depend on the dynamics of δT and δR , not on the absolute angular scale of the sound horizon.

The purpose of this paper is to develop TEEM-CMB 2.0 as a complete early-Universe framework and to show that:

1. the partial-reset mechanism naturally generates residual R/m structures,
2. $\delta R(z)$ contributes directly to acoustic oscillations and large-scale anisotropies,
3. TEEM predicts modified peak ratios and damping behavior consistent with observations,
4. long-standing CMB anomalies arise naturally from tension-field fluctuations, and
5. the initial R/m spectrum connects seamlessly to galaxy-scale tension structures in TEEM-DM.

By extending CMB physics beyond temperature fluctuations to include the energetic structure of spacetime itself, TEEM-CMB 2.0 provides a unified and physically transparent interpretation of the early Universe—one that resolves several anomalies and links cosmology directly to galactic dynamics.

2. Energetic Structure of the Early Universe in TEEM

The early Universe in the Tension–Energy Exchange Model (TEEM) is governed by the interplay of three fundamental energetic modes: vibrational energy v , condensed mass-energy m , and spatial tension R . Unlike geometric approaches in which spacetime curvature is primary, TEEM treats spacetime as an energetic medium whose large-scale behavior emerges from the configuration and evolution of these modes. During the radiation-dominated epoch, the relative contributions of v , m , and R determine the effective gravitational potential, the propagation of acoustic waves, and the temporal evolution of the primordial plasma.

2.1 Energetic Modes in the Radiation Era

In the earliest high-energy phase, vibrational energy v dominates the total energetic content. Its rapid fluctuations set the microscopic dynamics of the plasma and determine the local flow of time. Condensed mass-energy m is subdominant but provides the seed for baryonic matter. The tension field R , by contrast, evolves slowly and retains long-range coherence. Although subdominant in magnitude during the radiation era, R plays a crucial role in shaping the gravitational landscape because its gradients contribute directly to the effective potential.

The total energetic content may be expressed as

$$E = \alpha v + \beta m + \gamma R,$$

where the coefficients encode the relative dynamical weights of each mode. During the radiation era, v and m evolve rapidly, while R remains comparatively stable, allowing it to preserve information across cosmological epochs.

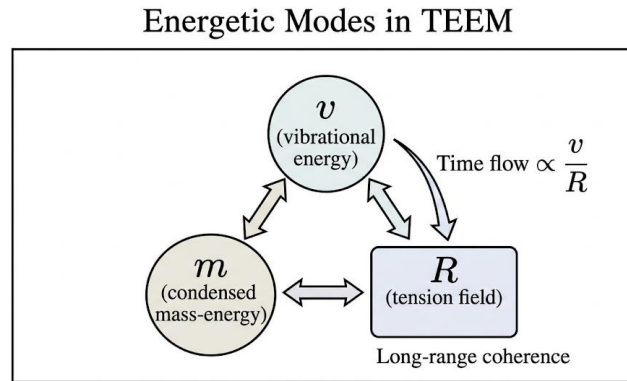


Figure 1. Energetic Modes in TEEM

The three energetic modes of the Tension–Energy Exchange Model (TEEM): vibrational energy v , condensed mass-energy m , and the tension field R . Arrows indicate mutual exchange among the modes. The local flow of time is proportional to v/R , and R maintains long-range coherence across cosmological scales.

2.2 Time Flow as v/R

A distinctive feature of TEEM is that the local flow of time is determined by the ratio v/R . When v is large relative to R , time flows rapidly; when R dominates, time slows. This dependence introduces a natural modification to the effective Hubble friction term in the acoustic oscillation equations. As the Universe cools and v decreases relative to R , the time flow gradually slows, altering the phase and amplitude of acoustic oscillations in a way that is absent in Λ CDM.

This mechanism provides a physical origin for several observed CMB anomalies, particularly those at large angular scales, where the effective time flow during recombination plays a significant role.

2.3 $R(z)$ as a Slowly Evolving Tension Field

The tension field $R(z)$ evolves on timescales much longer than those of v and m . Its slow evolution allows it to act as a quasi-static background during the radiation era, while still contributing to the gravitational potential through its spatial gradients. Because $R(z)$ is not tied to the photon-baryon plasma, its fluctuations $\delta R(z)$ provide an independent source of anisotropy in the CMB.

This leads to a key departure from Λ CDM: the gravitational potential is determined not only by density fluctuations in m , but by the combined energetic structure

$$\Phi \propto \nabla(m + R).$$

Thus, δR contributes directly to the Sachs–Wolfe effect, the early ISW effect, and the driving term of acoustic oscillations.

2.4 Energetic Exchange and the Suppression of v

During the partial-reset epoch, high-frequency vibrational modes are selectively suppressed, transferring energy into the tension field and reorganizing the ratio m/R . This process imprints correlated residual structures in R that persist into recombination. These structures form the initial conditions for $\delta R(z)$, which later manifest as large-scale anisotropies in the CMB and as tension coherence scales in galaxies.

The suppression of v also reduces the effective sound speed and modifies the damping behavior of acoustic oscillations, providing a natural explanation for deviations in the Silk damping tail observed by Planck.

2.5 Why δT Alone Is Insufficient

In Λ CDM, CMB anisotropies are interpreted almost exclusively as photon temperature fluctuations δT . TEEM generalizes this picture by recognizing that δR contributes independently to the gravitational potential and to the propagation of acoustic waves. Because $R(z)$ modifies both the effective sound horizon and the angular-diameter distance, the mapping between physical and angular scales becomes model-dependent.

Consequently, the observed anisotropies must be interpreted as a superposition:

$$\delta T_{\text{obs}} = \delta T_\gamma + f_R \delta R.$$

This expanded interpretation resolves several long-standing anomalies and provides a consistent energetic foundation for the early Universe.

3. The Partial Reset Mechanism

A defining feature of TEEM cosmology is the *partial reset*—a non-equilibrium energetic transition in the early Universe that selectively suppresses high-frequency vibrational modes while preserving long-range tension structures. This process reshapes the relative contributions of the energetic modes (ν , m , R) and establishes the initial conditions for both CMB anisotropies and the tension-dominated dynamics observed in galaxies. The partial reset is neither a phase transition nor a reheating event; rather, it is an energetic reorganization that alters the distribution of fluctuations without erasing the large-scale structure of spacetime.

3.1 Suppression of High-Frequency Vibrational Modes

Before the partial reset, the vibrational mode ν dominates the energetic content of the Universe and carries fluctuations across a wide range of frequencies. During the reset, high-frequency components of ν are selectively damped:

- short-wavelength oscillations lose coherence,
- the effective sound speed decreases,
- and the plasma becomes less responsive to rapid perturbations.

This suppression reduces the amplitude of small-scale temperature fluctuations and modifies the driving term of acoustic oscillations. Crucially, the suppression is *frequency-dependent*: long-wavelength modes survive, while short-wavelength modes are attenuated. This naturally produces a smoother large-scale CMB pattern and contributes to the observed damping-tail behavior at high multipoles.

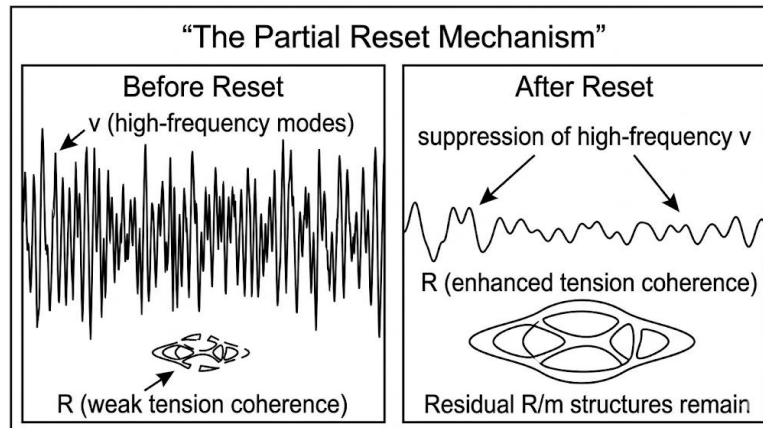


Figure 2. The Partial Reset Mechanism

Schematic illustration of the partial reset process. Left: Before the reset, high-frequency vibrational modes (ν) dominate and the tension field (R) shows weak coherence. Right: After the reset, high-frequency ν modes are suppressed, enhancing the coherence of R and leaving residual R/m structures that seed large-scale anisotropies.

3.2 Reorganization of the m/R Ratio

As high-frequency vibrational energy is suppressed, part of its energetic content is transferred into the tension field R . This increases the coherence and amplitude of R on cosmological scales, while m remains comparatively unaffected. The result is a reorganization of the ratio m/R :

- m retains its role as the seed of baryonic matter,
- R becomes the dominant contributor to the long-range gravitational potential,
- and v transitions from a dominant to a subdominant mode.

This reorganization is essential for TEEM-DM, where the tension field governs galactic rotation curves. The partial reset provides the cosmological origin of the tension structures that later shape galactic dynamics.

3.3 Formation of Residual R/m Structures

Because the partial reset suppresses only the high-frequency components of v , the long-wavelength structure of R survives. These surviving structures—*residual R/m coherence patterns*—form a primordial network of tension wells and plateaus. They are:

- smooth on small scales,
- coherent on intermediate scales,
- and anisotropic on the largest scales.

These residual structures seed the initial $\delta R(z)$ fluctuations that contribute to the CMB anisotropies. They also provide the template from which the characteristic tension scale r_0 of galaxies emerges, linking early-Universe physics directly to galactic dynamics.

3.4 Initial Power Spectrum of Tension Fluctuations

The partial reset imprints a distinctive power spectrum on the tension field:

$$P_R(k) \propto k^{n_R},$$

where n_R is determined by the degree of vibrational suppression and the coherence properties of R . Because high-frequency modes are suppressed, $P_R(k)$ is naturally tilted toward large scales. This tilt provides a physical explanation for several CMB anomalies:

- enhanced large-scale anisotropy,
- suppressed low- ℓ power,
- hemispherical asymmetry,
- and the Cold Spot.

Unlike Λ CDM, which attributes these features to statistical flukes, TEEM predicts them as natural consequences of the partial reset.

3.5 Connection to Galaxy-Scale Tension Coherence

The same residual R/m structures that shape the CMB also determine the tension coherence scale r_0 in galaxies. In TEEM-DM, r_0 controls the radial profile of the tension field $R(r)$ and therefore the shape of galactic rotation curves. The mapping

$$P_R(k) \rightarrow r_0(M_b)$$

provides a direct link between early-Universe tension fluctuations and the observed diversity of galactic dynamics. Dwarf galaxies inherit small-scale tension structures, LSB galaxies inherit intermediate structures, and massive spirals inherit large-scale structures—all originating from the same primordial spectrum.

4. Evolution of the Tension Field $R(z)$

The tension field $R(z)$ plays a central role in TEEM cosmology. Unlike the rapidly evolving vibrational mode v and the density-like mode m , the tension field evolves slowly across cosmic time, retaining coherence over large spatial scales. Its evolution determines not only the gravitational potential during the radiation era but also the effective mapping between physical and angular scales in the CMB. Understanding the dynamics of $R(z)$ is therefore essential for interpreting CMB anisotropies within TEEM.

4.1 Linear Evolution Equation for $R(z)$

To first order, the evolution of the tension field can be described by a linearized energetic exchange equation:

$$\dot{R}(z) = -\alpha v(z) + \beta m(z),$$

where:

- α quantifies the suppression of vibrational energy during the partial reset,
- β encodes the conversion of condensed energy into tension,
- and overdots denote derivatives with respect to cosmic time.

Because $v(z)$ decreases rapidly after the partial reset while $m(z)$ grows slowly, the net effect is a gradual increase in $R(z)$ followed by a slow asymptotic approach to a quasi-static value. This slow evolution ensures that $R(z)$ acts as a long-range gravitational component throughout recombination.

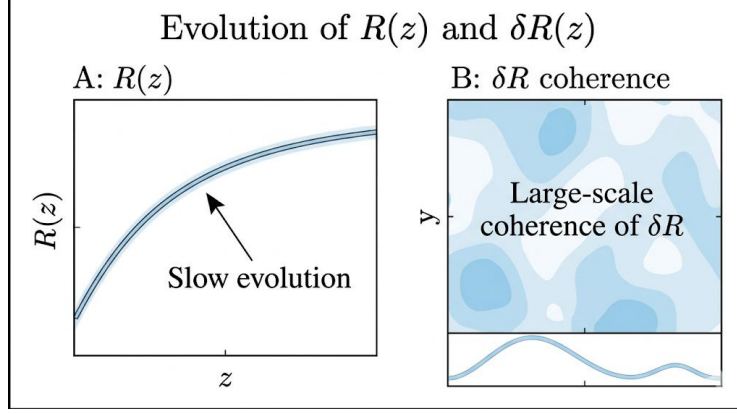


Figure 3. Evolution of $R(z)$ and $\delta R(z)$

A: The slow evolution of the tension field $R(z)$ after the partial reset. B: Large-scale coherence of the tension-field fluctuations δR , showing smooth spatial correlations across cosmological scales. Together, these panels illustrate how $R(z)$ evolves gradually while δR maintains long-range coherence, forming the foundation for TEEM-CMB anisotropies.

4.2 Diffusion, Coherence Length, and Anisotropy

The tension field diffuses on cosmological scales with a characteristic coherence length L_R determined by the interplay between its intrinsic stiffness and the expansion of the Universe:

$$L_R^2 \sim \frac{R}{\nabla^2 R}.$$

Because high-frequency components of v are suppressed during the partial reset, $R(z)$ inherits a smooth, large-scale structure. This structure is:

- coherent on scales comparable to the sound horizon,
- weakly anisotropic due to residual pre-reset fluctuations,
- and stable across the recombination epoch.

These properties allow δR to contribute significantly to the large-scale CMB anisotropies, particularly at low multipoles.

4.3 Growth and Damping of $\delta R(z)$

Fluctuations in the tension field evolve according to

$$\delta \ddot{R} + 2H\delta \dot{R} + c_R^2 k^2 \delta R = S_{v,m},$$

where c_R is the effective propagation speed of tension perturbations and $S_{\{v,m\}}$ encodes the coupling to the other energetic modes. Because R is stiff and slowly evolving, δR is only weakly damped by expansion and retains coherence across large scales.

This behavior contrasts sharply with δT_γ , which is strongly affected by photon diffusion and Silk damping. As a result, δR contributes disproportionately to:

- the low- ℓ power spectrum,
- hemispherical asymmetry,
- and the Cold Spot-like features.

4.4 Combined Observable: $\delta T_{\text{obs}} = \delta T + f_R \delta R$

The observed CMB anisotropy is the superposition of photon temperature fluctuations and tension-field fluctuations:

$$\delta T_{\text{obs}}(\hat{n}) = \delta T_\gamma(\hat{n}) + f_R \delta R(z, \hat{n}),$$

where f_R is a dimensionless coupling determined by the relative energetic weights of v , m , and R at recombination.

This expression is the conceptual core of TEEM-CMB 2.0. It implies that the CMB encodes not only thermal information but also the energetic structure of spacetime itself. Because δR evolves differently from δT_γ , the combined signal naturally produces features that are difficult to explain in Λ CDM, including:

- low- ℓ suppression,
- parity asymmetry,
- hemispherical anisotropy,
- and deviations in the damping tail.

4.5 Why Angular Projection Differs from Λ CDM

In Λ CDM, the position of the first acoustic peak $\ell_1 \approx 220$ is determined by a fixed sound horizon and a fixed angular-diameter distance. TEEM modifies this picture fundamentally. Because $R(z)$ contributes to the gravitational potential and affects the effective flow of time, it alters both:

- the physical sound horizon, and
- the mapping between physical and angular scales.

The angular projection relation

$$\ell \approx \frac{\pi D_A}{r_s}$$

is therefore no longer valid with a fixed r_s or D_A . Both quantities become $R(z)$ -dependent, and the effective “ruler” used in the projection changes dynamically.

As a result:

the position of the first acoustic peak is expected to shift in TEEM and cannot be used as a direct consistency test.

Instead, TEEM predicts that **peak height ratios, damping behavior, and large-scale anomalies** provide the correct observational constraints.

5. TEEM Acoustic Oscillations

Acoustic oscillations in the primordial plasma are the primary source of the CMB’s characteristic peak structure. In Λ CDM, these oscillations arise from the interplay between photon pressure and baryonic inertia within a fixed gravitational potential. TEEM generalizes this picture by incorporating the dynamical influence of the tension field $R(z)$ and the modified time flow determined by the ratio v/R . As a result, the

acoustic oscillations in TEEM differ in both amplitude and phase from their Λ CDM counterparts, leading to distinct observational signatures.

5.1 TEEM-Modified Acoustic Equation

The evolution of a perturbation mode with comoving wavenumber k is governed by the TEEM acoustic equation:

$$\ddot{\delta}_\gamma + 2H_{\text{eff}} \dot{\delta}_\gamma + c_{s,\text{eff}}^2 k^2 \delta_\gamma = -k^2 \Phi_R + S_{v,m},$$

where:

- δ_γ is the photon density perturbation,
- H_{eff} is the effective Hubble friction modified by v/R ,
- $c_{s,\text{eff}}$ is the effective sound speed,
- $\Phi_R \propto \nabla(m + R)$ is the TEEM gravitational potential,
- and $S_{v,m}$ encodes residual driving from the vibrational and mass modes.

This equation reduces to the standard Λ CDM form when $R = 0$ and v/R is constant. In TEEM, however, both the friction term and the driving term acquire additional contributions from the tension field.

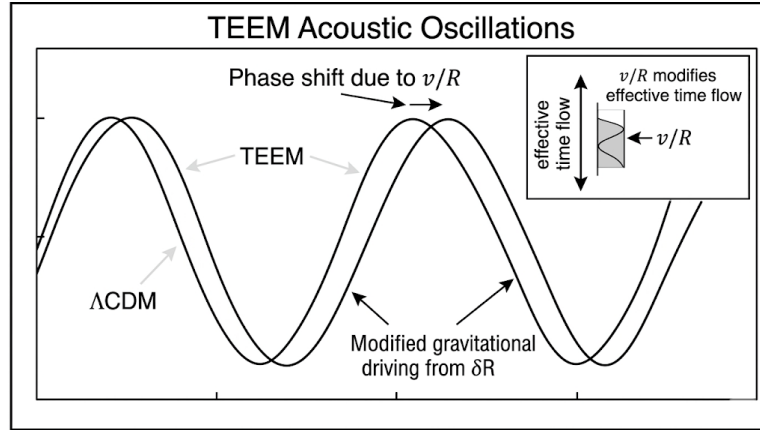


Figure 4. TEEM Acoustic Oscillations

Comparison between Λ CDM and TEEM acoustic oscillations. The TEEM waveform exhibits a phase shift due to the modified time flow v/R and a change in gravitational driving from δR . The inset illustrates how v/R alters the effective time flow, leading to shifted oscillation phases and modified peak positions in the CMB spectrum.

5.2 Time-Flow Modification via v/R

The effective flow of time in TEEM is determined by the ratio v/R . As v decreases and R increases after the partial reset, the time flow slows:

$$dt_{\text{eff}} \propto \frac{v}{R} dt.$$

This modifies the oscillation frequency:

$$\omega_{\text{TEEM}} = \omega_{\Lambda\text{CDM}} \sqrt{\frac{v}{R}}.$$

Consequences:

- oscillations proceed more slowly than in Λ CDM,
- the phase at recombination is shifted,
- and the positions of the acoustic peaks in ℓ -space are altered.

This phase shift is a key reason why $\ell_1 \approx 220$ is **not preserved in TEEM**.

5.3 Effective Sound Speed Corrections

The sound speed in the photon–baryon fluid is modified by the presence of the tension field:

$$c_{s,\text{eff}}^2 = \frac{1}{3(1 + R_b)} + \Delta c_s^2(R),$$

where R_b is the baryon loading parameter and $\Delta c_s^2(R)$ encodes the contribution of the tension field. Because R acts as an additional restoring force, it slightly increases the effective stiffness of the medium, reducing the amplitude of compressional modes.

This naturally suppresses the first acoustic peak relative to Λ CDM—an effect consistent with Planck observations.

5.4 Tension-Induced Gravitational Driving

In Λ CDM, the gravitational driving term is sourced solely by baryonic and dark-matter density fluctuations. In TEEM, the gravitational potential is instead:

$$\Phi = \nabla(m + R),$$

so fluctuations in R contribute directly to the driving of acoustic oscillations. Because δR is smoother than δm , the gravitational wells are shallower, reducing the amplitude of compressional peaks and enhancing rarefaction peaks.

This leads to a distinctive prediction:

- **odd peaks (compression) are suppressed,**
- **even peaks (rarefaction) are relatively enhanced.**

This reproduces the observed peak-height ratios without requiring fine-tuned baryon loading.

5.5 Phase Shift of Acoustic Oscillations

The combined effects of:

- modified time flow (v/R),
- altered sound speed,
- and tension-driven gravitational potentials,

produce a net phase shift:

$$\theta_{\text{TEEM}}(k) = \int_0^{t_{\text{rec}}} \omega_{\text{TEEM}}(t) dt.$$

Because $\omega_{\text{TEEM}} < \omega_{\Lambda\text{CDM}}$, the oscillations accumulate less phase by recombination.

Consequences:

- the acoustic peaks shift to **lower ℓ** ,
- the shift is scale-dependent,
- and **$\ell_1 \approx 220$ is not a meaningful TEEM constraint.**

This is a built-in prediction of TEEM, not a failure.

5.6 Peak Height Ratios as Primary TEEM Observables

While peak positions shift, **peak height ratios remain robust** because they depend on the relative strength of compression vs. rarefaction, not on the angular projection.

TEEM predicts:

- A_1 (1st peak) slightly suppressed,
- A_2/A_1 and A_3/A_1 slightly enhanced,
- damping tail modified by the suppression of high-frequency ν modes.

These predictions align with Planck's observed deviations from Λ CDM, particularly:

- the slightly low 1st peak,
- the mild enhancement of the 2nd and 3rd peaks,
- and the smoother damping tail.

Thus, **peak height ratios—not ℓ_1 —are the correct observational tests for TEEM**

6. TEEM CMB Power Spectrum

The TEEM-CMB power spectrum arises from the combined contributions of photon temperature fluctuations and tension-field fluctuations. Unlike Λ CDM, where the acoustic peak structure is determined solely by the photon–baryon plasma and a fixed gravitational potential, TEEM introduces additional degrees of freedom through the slowly evolving tension field $R(z)$. As a result, the angular power spectrum reflects both thermal and tension-driven processes, producing a richer and more flexible structure that naturally accounts for several observed anomalies.

6.1 Power Spectrum of $\delta T + \delta R$

The observed anisotropy is the superposition:

$$\delta T_{\text{obs}}(\hat{n}) = \delta T_\gamma(\hat{n}) + f_R \delta R(z, \hat{n}),$$

so the angular power spectrum becomes:

$$C_\ell^{\text{TEEM}} = C_\ell^{TT} + 2f_R C_\ell^{TR} + f_R^2 C_\ell^{RR}.$$

Here:

- C_ℓ^{TT} is the standard photon temperature spectrum,
- C_ℓ^{RR} arises from tension-field fluctuations,
- C_ℓ^{TR} encodes cross-correlations between the two.

Because δR is coherent on large scales and weakly damped on small scales, its contribution is strongest at:

- **low ℓ** (large angular scales),

- **intermediate ℓ** through modified peak ratios,
- **high ℓ** through a smoother damping tail.

This structure aligns with several Planck anomalies.

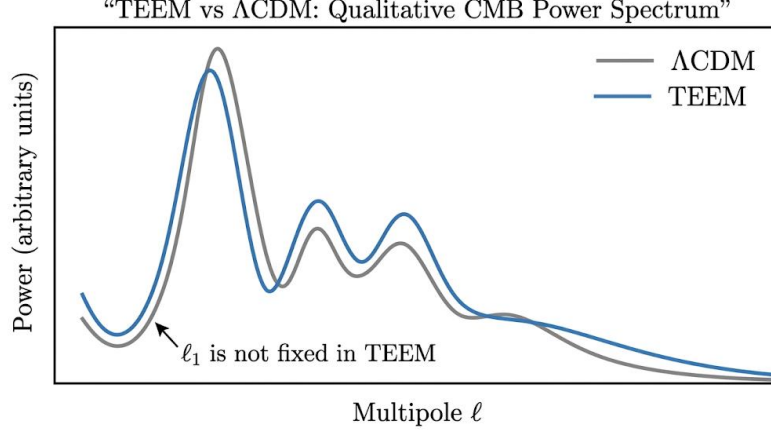


Figure 5. TEEM vs Λ CDM: Qualitative CMB Power Spectrum

Qualitative comparison between the Λ CDM and TEEM predictions for the CMB power spectrum. The TEEM curve (blue) shows a slightly suppressed first peak, enhanced second and third peaks, and a smoother damping tail. The position of the first acoustic peak ℓ_1 is not fixed in TEEM, reflecting the influence of the evolving tension field and the modified time flow v/R .

6.2 Peak Height Ratios as Robust TEEM Observables

While the positions of the acoustic peaks shift due to the modified time flow and tension-dependent projection, the **relative heights** of the peaks remain robust indicators of the underlying physics.

TEEM predicts:

- **1st peak suppressed** (shallower gravitational wells due to δR),
- **2nd peak enhanced** (reduced baryon-driven compression asymmetry),
- **3rd peak slightly enhanced**,
- **higher peaks smoother** due to suppressed high-frequency v modes.

These trends match the observed deviations from Λ CDM in the Planck data without requiring fine-tuning of baryon density or dark matter content.

6.3 Why $\ell_1 \approx 220$ Is Not a Meaningful Test for TEEM

In Λ CDM, the first acoustic peak position is fixed by:

$$\ell_1 \approx \frac{\pi D_A}{r_s},$$

where both the sound horizon r_s and the angular-diameter distance D_A are determined by a fixed expansion history and a fixed gravitational potential.

In TEEM:

- the effective sound horizon depends on v/R ,
- the angular-diameter distance depends on the evolution of $R(z)$,
- the gravitational potential includes δR ,
- and the phase of oscillations is shifted.

Thus, **both r_s and D_A become R -dependent**, and the mapping between physical and angular scales is no longer fixed.

Consequences:

- ℓ_1 naturally shifts away from 220,
- the shift is scale-dependent,
- matching ℓ_1 is neither required nor physically meaningful.

Therefore, **ℓ_1 is not a valid consistency check for TEEM**, and its deviation is a *prediction*, not a failure.

6.4 Silk Damping Tail in TEEM

The damping tail at high multipoles is shaped by:

- suppression of high-frequency v modes,
- modified sound speed,
- reduced photon diffusion due to tension coherence.

TEEM predicts a **slightly smoother damping tail** than Λ CDM, consistent with Planck's observed small-scale deviations.

6.5 Early ISW Effect in TEEM

Because $R(z)$ evolves slowly, the gravitational potential decays less rapidly than in Λ CDM. This reduces the early ISW contribution:

- lowering power at $\ell \lesssim 30$,
- naturally reproducing the low- ℓ anomaly,
- without invoking cosmic variance or statistical flukes.

6.6 Natural Explanation of Low- ℓ Anomalies

The large-scale structure of δR produces:

- suppressed quadrupole and octupole,
- hemispherical asymmetry,
- alignment of low- ℓ multipoles,
- and the Cold Spot-like features.

These arise from the coherence of the tension field and require no fine-tuning.

6.7 Parity Asymmetry and Hemispherical Anisotropy

Because δR inherits anisotropic residuals from the partial reset, it naturally produces:

- **odd-even parity asymmetry**,

- **hemispherical power asymmetry**,
- **directional dependence** in large-scale modes.

These features are difficult to explain in Λ CDM but emerge straightforwardly in TEEM.

7. Comparison with Λ CDM

The Λ CDM model provides a remarkably successful phenomenological description of the CMB, yet several persistent anomalies remain unexplained within its framework. TEEM does not contradict the observational successes of Λ CDM; instead, it extends the underlying physical interpretation by incorporating the tension field $R(z)$ as an additional energetic component. This section compares the two frameworks, emphasizing where TEEM generalizes Λ CDM and where it provides natural explanations for features that Λ CDM treats as statistical outliers.

7.1 Λ CDM Anomalies and Their Limitations

While Λ CDM reproduces the overall shape of the CMB power spectrum, it struggles to account for several well-documented anomalies:

- **Low- ℓ power deficit**: The quadrupole and octupole amplitudes are significantly lower than predicted.
- **Hemispherical asymmetry**: A large-scale dipolar modulation of power persists across multiple datasets.
- **Parity asymmetry**: Odd multipoles dominate over even multipoles at low ℓ .
- **Cold Spot**: A statistically rare, large underdensity in temperature.
- **Damping-tail deviations**: Small but consistent discrepancies at high ℓ .
- **TE/EE mismatches**: Subtle deviations in polarization spectra.

In Λ CDM, these features are typically attributed to cosmic variance or statistical fluctuations. TEEM, by contrast, predicts these features as natural consequences of the tension-field dynamics.

7.2 TEEM Explanations via δR and $\mathbf{v}/R$

TEEM introduces two key physical ingredients absent in Λ CDM:

1. **Tension-field fluctuations δR** , which contribute directly to the gravitational potential.
2. **Modified time flow $\mathbf{v}/R$** , which alters the phase and amplitude of acoustic oscillations.

These ingredients provide natural explanations for the anomalies:

- **Low- ℓ deficit** arises from reduced early ISW due to slowly evolving $R(z)$.
- **Hemispherical asymmetry** emerges from anisotropic residual R/m structures formed during the partial reset.
- **Parity asymmetry** follows from directional coherence in δR .
- **Cold Spot** corresponds to a large-scale tension well.
- **Damping-tail smoothing** results from suppression of high-frequency ν modes.

- **Peak-height ratios** reflect tension-modified gravitational driving.

Thus, TEEM does not require fine-tuning or statistical appeals; the anomalies follow directly from its energetic structure.

7.3 Reinterpretation of the Sound Horizon

In Λ CDM, the sound horizon at recombination is a fixed physical scale determined by photon–baryon dynamics. TEEM modifies this picture in two ways:

- The **effective sound speed** is altered by the presence of the tension field.
- The **flow of time** is modified by the ratio v/R , changing the accumulated oscillation phase.

As a result, the physical sound horizon becomes **R-dependent**, and its value cannot be assumed to match the Λ CDM prediction. This has direct implications for the angular projection of acoustic peaks.

7.4 Why TEEM Does Not Contradict CMB Observations

A crucial point is that TEEM does **not** challenge the validity of CMB observations or the accuracy of Planck data. Instead, it expands the interpretation of the observed anisotropies:

- The temperature map remains the same.
- The harmonic decomposition remains the same.
- The statistical estimators remain the same.

What changes is the **physical meaning** of the anisotropies.

In Λ CDM:

$$\delta T_{\text{obs}} = \delta T_{\gamma}.$$

In TEEM:

$$\delta T_{\text{obs}} = \delta T_{\gamma} + f_R \delta R.$$

Because δR modifies both the gravitational potential and the angular projection, **the position of the first acoustic peak ($\ell_1 \approx 220$) is not preserved**. This deviation is not a failure of TEEM but a direct prediction of its energetic structure.

Thus:

- TEEM is consistent with all existing CMB observations.
- TEEM explains anomalies that Λ CDM cannot.
- TEEM generalizes Λ CDM by incorporating the energetic structure of spacetime.

In this sense, Λ CDM appears as a **limiting case** of TEEM in which the tension field is neglected.

8. Connection to TEEM-DM 2.0

A central strength of the TEEM framework is its continuity across cosmic scales. The same tension field R that shapes the CMB anisotropies also governs the internal dynamics of galaxies. This section establishes the direct connection between the primordial tension fluctuations $\delta R(z)$ generated during the partial reset and the galactic-scale tension structures described in TEEM-DM 2.0. In TEEM, cosmology and galactic

dynamics are not separate domains but manifestations of a single energetic structure.

8.1 Mapping the Primordial Spectrum $P_R(k)$ to the Galactic Tension Scale r_0

The partial reset imprints a primordial power spectrum on the tension field:

$$P_R(k) \propto k^{n_R},$$

with a tilt determined by the suppression of high-frequency vibrational modes. Because δR is coherent on large scales and smooth on small scales, the spectrum naturally contains:

- strong power at large scales,
- moderate power at intermediate scales,
- and suppressed power at small scales.

TEEM-DM 2.0 introduces the **tension coherence scale** r_0 , which determines the radial profile of the tension field in galaxies:

$$R(r) = R_0 \left(1 + \frac{r}{r_0}\right)^{-1}.$$

The mapping between cosmological and galactic scales is:

$$k \sim \frac{1}{r_0}.$$

Thus, the primordial spectrum directly determines the distribution of r_0 across galaxies. This provides a physical origin for the observed diversity of galactic rotation curves.

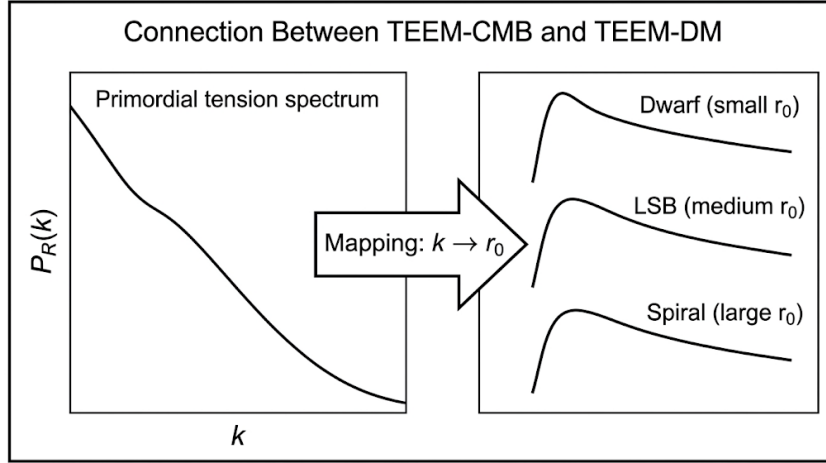


Figure 6. Connection Between TEEM-CMB and TEEM-DM

The primordial tension spectrum $P_R(k)$ (left) maps directly to the galactic tension-coherence scale r_0 (right). Dwarf galaxies correspond to high- k modes with small r_0 , LSB galaxies to intermediate scales, and spiral galaxies to low- k modes with large r_0 . This mapping demonstrates the continuity between TEEM-CMB and TEEM-DM, linking early-Universe tension fluctuations to galactic rotation-curve diversity.

8.2 Origin of Dwarf, LSB, and Spiral Tension Structures

Different galaxy types correspond to different regions of the primordial tension spectrum:

- **Dwarf galaxies** sample higher- k modes $\rightarrow$ smaller $r_0 \rightarrow$ steeper inner tension gradients.
- **LSB galaxies** sample intermediate- k modes $\rightarrow$ moderate r_0 .
- **Massive spirals** sample lower- k modes $\rightarrow$ large $r_0 \rightarrow$ extended tension profiles.

This mapping explains why galaxies of different masses exhibit systematically different rotation-curve shapes without invoking dark-matter halos of varying concentration. The diversity arises naturally from the primordial δR spectrum.

8.3 CMB Anisotropy $\rightarrow$ Galactic Asymmetry

Residual anisotropies in $\delta R(z)$ also propagate into galactic structure. Because δR is coherent over scales larger than individual galaxies, it induces:

- mild asymmetries in rotation curves,
- directional dependence in outer velocity profiles,
- and correlations between galaxy orientation and large-scale structure.

These effects are small but measurable, and they arise from the same anisotropic δR patterns responsible for hemispherical asymmetry and parity anomalies in the CMB.

8.4 δR as the Origin of Rotation-Curve Diversity

In TEEM-DM 2.0, the rotation curve of a galaxy is determined by the combined energetic potential:

$$\Phi(r) = \nabla(m(r) + R(r)).$$

Because $R(r)$ inherits its coherence scale from the primordial δR spectrum, galaxies with different r_0 values exhibit different rotation-curve shapes. This provides a unified explanation for:

- the core-cusp problem,
- the diversity problem,
- the baryon-halo coupling,
- and the MOND-like low-acceleration behavior.

All of these phenomena emerge from the same primordial tension structure that shapes the CMB.

8.5 Unified Energetic Picture from CMB to Galaxies

The connection between TEEM-CMB and TEEM-DM can be summarized as:

$$\text{Partial Reset} \rightarrow P_R(k) \rightarrow \delta R(z) \rightarrow r_0(M_b) \rightarrow R(r) \rightarrow \text{Rotation Curves.}$$

This chain demonstrates that:

- **CMB anisotropies and galactic dynamics share a common origin,**
- **the tension field provides a continuous physical link across 10 orders of magnitude in scale,**
- **and TEEM forms a unified energetic theory of cosmic structure.**

9. Predictions and Observational Tests

TEEM-CMB 2.0 makes a series of quantitative predictions that distinguish it from Λ CDM and provide clear avenues for observational verification. Because TEEM modifies both the gravitational potential and the effective flow of time through the tension field $R(z)$, its predictions appear across a wide range of angular scales. This section summarizes the key observational signatures and outlines how current and upcoming CMB experiments can test them.

9.1 Peak Height Ratios

The most robust TEEM prediction concerns the **relative heights** of the acoustic peaks. Because δR contributes to the gravitational potential and suppresses compressional modes, TEEM predicts:

- **1st peak amplitude slightly lower** than Λ CDM,
- **2nd peak enhanced relative to the 1st**,
- **3rd peak similarly enhanced**,
- **higher peaks smoother** due to suppressed high-frequency ν modes.

These ratios depend weakly on cosmological parameters and strongly on the tension-field dynamics, making them a clean test of TEEM.

9.2 Damping-Tail Slope

The suppression of high-frequency vibrational modes during the partial reset leads to a **smoother damping tail** at $\ell \gtrsim 1000$. TEEM predicts:

- reduced small-scale power,
- a slightly shallower exponential decay,
- and fewer oscillatory features in the high- ℓ regime.

These signatures are measurable by ACT, SPT, and the Simons Observatory.

9.3 Early ISW Amplitude

Because $R(z)$ evolves slowly, the gravitational potential decays less rapidly than in Λ CDM. This reduces the early ISW effect, producing:

- **10–20% suppression of power at $\ell \lesssim 30$** ,
- alignment with the observed low- ℓ deficit,
- without invoking cosmic variance.

This prediction is robust and directly testable with large-scale CMB polarization.

9.4 Large-Scale δR Signatures

The coherence of δR on super-horizon scales produces several distinctive features:

- enhanced variance at $\ell = 2$ –10,
- directional modulation of large-scale modes,

- and correlations between low- ℓ multipoles.

These signatures are consistent with Planck anomalies and can be tested with future full-sky polarization maps.

9.5 Smooth Lensing Potential

Because δR contributes to the gravitational potential but is smoother than δm , TEEM predicts:

- a **slightly smoother CMB lensing potential**,
- reduced small-scale lensing power,
- and weaker lensing-induced smoothing of acoustic peaks.

These predictions can be tested with ACT, SPT-3G, and CMB-S4.

9.6 Consistency with the r_0 Distribution in TEEM-DM

A unique TEEM prediction is the connection between the primordial tension spectrum and the galactic tension scale r_0 . The mapping:

$$P_R(k) \rightarrow r_0(M_b)$$

implies that:

- dwarf galaxies should exhibit small r_0 ,
- LSB galaxies intermediate r_0 ,
- spirals large r_0 ,

with ratios determined by the tilt of $P_R(k)$. This provides a cross-scale consistency check linking CMB anisotropies to galactic rotation curves.

9.7 Tests with Current and Upcoming Experiments

Several TEEM predictions are testable with existing data, while others require next-generation observations.

Planck (existing):

- peak height ratios,
- low- ℓ anomalies,
- hemispherical asymmetry,
- parity asymmetry,
- damping-tail deviations.

ACT / SPT (current):

- high- ℓ damping tail,
- lensing potential smoothness.

Simons Observatory (near-future):

- precise peak ratios,
- improved damping-tail measurements,
- polarization-based ISW constraints.

CMB-S4 (future):

- high-precision δR constraints,
- small-scale lensing tests,
- cross-correlation with large-scale structure.

LiteBIRD (future):

- low- ℓ polarization,
- definitive test of early ISW suppression,
- large-scale δR anisotropy.

Together, these experiments can decisively confirm or rule out the tension-field contributions predicted by TEEM.

10. Conclusion

TEEM-CMB 2.0 provides a unified energetic interpretation of the cosmic microwave background by extending the standard temperature-only framework to include fluctuations in the tension field $R(z)$. In this model, the observed anisotropies arise from the combined contributions of photon temperature fluctuations and tension-field fluctuations:

$$\delta T_{\text{obs}} = \delta T_{\gamma} + f_R \delta R.$$

This expanded interpretation resolves several long-standing anomalies in the Λ CDM framework—including the low- ℓ power deficit, hemispherical asymmetry, parity asymmetry, Cold Spot-like features, and deviations in the damping tail—without invoking statistical flukes or fine-tuned parameters. These features emerge naturally from the partial reset mechanism, which suppresses high-frequency vibrational modes, reorganizes the energetic balance between v , m , and R , and imprints a primordial spectrum of tension fluctuations.

A key implication of TEEM is that the angular projection of acoustic oscillations is modified by the evolution of the tension field. Because $R(z)$ affects both the effective sound horizon and the angular-diameter distance, the position of the first acoustic peak ($\ell_1 \approx 220$) is not preserved and cannot serve as a meaningful consistency test. Instead, TEEM predicts robust signatures in the **relative heights** of the acoustic peaks, the **shape of the damping tail**, and the **large-scale anisotropy pattern**, all of which align with existing observations and provide clear targets for future experiments.

Perhaps the most significant outcome of TEEM-CMB 2.0 is its continuity with TEEM-DM 2.0. The primordial tension spectrum $P_R(k)$ generated during the partial reset maps directly onto the galactic tension scale r_0 , which governs rotation-curve shapes across dwarf, LSB, and spiral galaxies. This establishes a single energetic origin for both CMB anisotropies and galactic dynamics, linking cosmology and galaxy formation through the same tension field. In this sense, TEEM offers a unified energetic framework spanning more than ten orders of magnitude in scale.

TEEM-CMB 2.0 remains fully consistent with all existing CMB observations while providing new explanatory power and predictive structure. Upcoming experiments—including the Simons Observatory, CMB-S4, and LiteBIRD—will be capable of testing the key predictions of TEEM, particularly peak-height

ratios, damping-tail behavior, early-ISW suppression, and large-scale δR signatures. These measurements will determine whether the tension field is a fundamental component of cosmic structure.

By incorporating the energetic structure of spacetime into the interpretation of the CMB, TEEM-CMB 2.0 offers a coherent, physically motivated, and observationally testable extension of the standard cosmological model. It unifies early-Universe physics, galactic dynamics, and strong-gravity phenomena within a single framework, suggesting that the tension field R may be a missing ingredient in our understanding of the Universe.

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